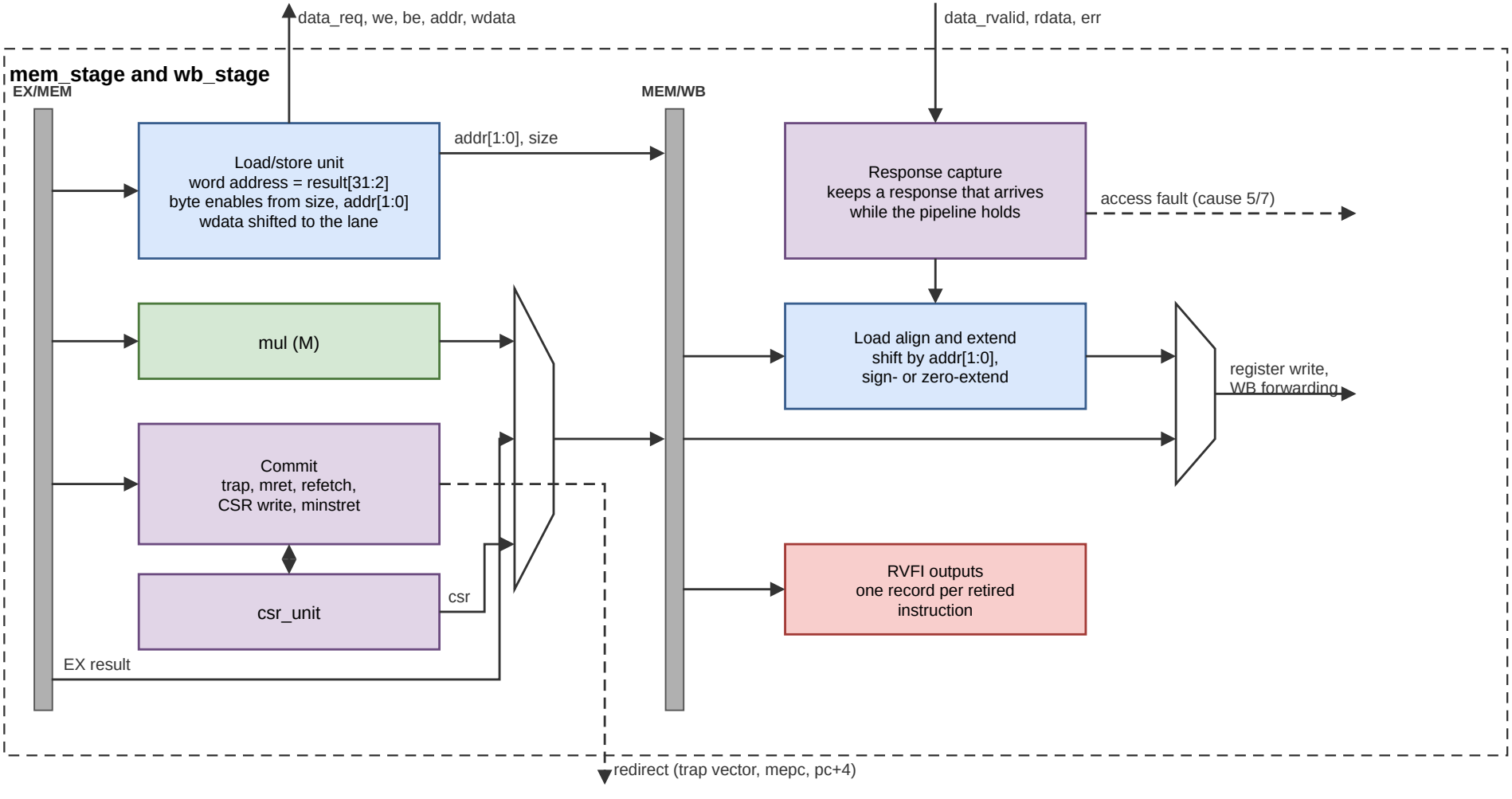
data_req, we, be, addr, wdata
data_rvalid, rdata, err
mem_stage and wb_stage
EX/MEM
MEM/WB
Load/store unit
word address = result[31:2]
byte enables from size, addr[1:0]
wdata shifted to the lane
addr[1:0], size
Response capture
keeps a response that arrives
while the pipeline holds
access fault (cause 5/7)
mul (M)
Load align and extend
shift by addr[1:0],
sign- or zero-extend
register write,
WB forwarding
Commit
trap, mret, refetch,
CSR write, minstret
csr_unit
csr
RVFI outputs
one record per retired
instruction
EX result
redirect (trap vector, mepc, pc+4)